Zohaib Ali Mughal

Full-Stack Software Engineer

z.mughal.24648@gmail.com | Karachi, Pakistan
career.zohaibmughal.dev | github.com/Zohaib24648 | linkedin.com/in/zohaibmughal24648

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Dart, SQL, Python
Frontend: Next.js, React, Tailwind CSS, TanStack Query, Radix / shadcn UI
Mobile: Flutter (BLoC, clean architecture), Capacitor, Firebase Cloud Messaging, App Store and Play Store releases
Backend: Node.js, NestJS, Fastify, tRPC, REST / OpenAPI, Zod, BullMQ, NATS JetStream
Data and DevOps: PostgreSQL, Prisma, Supabase, Redis, Docker, AWS, Railway, Vercel, GitHub Actions
Payments and Quality: Stripe Connect, Paymob, Airwallex, Sentry, Jest, Vitest, Playwright, Testcontainers
AI-assisted development: Claude Code, Cursor, GitHub Copilot

PROFESSIONAL EXPERIENCE

Software Engineer II, Slidey LLC, Karachi **Nov 2024 – Present**
Promoted twice in 18 months: Trainee (Nov 2024), Software Engineer I (Mar 2025), Software Engineer II (May 2026)

ETG Platform evolutiongroup.net.au | live portal | case study **Software Engineer II**

Field-service platform for an Australian electrical, security and telecom group: admin, client, supplier and technician portals. Built to stop about $5k a week lost to unbilled supplier costs and billing scheduled rather than actual hours.

- Designed the monorepo and its data model: a Next.js frontend, a Fastify and tRPC API and a BullMQ worker over one PostgreSQL schema for dispatch, timesheets, assets, quoting and billing.
- Enforcement-first decision engine: one pure, unit-tested gate checks every dispatch, closeout and invoice step and logs each block and manager override; a job cannot be invoiced until its supplier costs match the bank feed.
- Shared inbox and AI bill reading over Gmail: every email links to its job, quote or invoice, and supplier bills are parsed by per-supplier PDF readers or prefilled through the Claude API for a person to approve.
- Integration layer: typed adapters for more than ten external systems, including trade supplier portals with no public API; supplier bills are paid through Airwallex transfers with a second approver on larger payments.

Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, BullMQ, Redis

Hyss hyss.io | case study **Software Engineer I**

Live-event staffing marketplace: companies hire certified freelance crew, with competency-matched shifts, call sheets, realtime chat, invoicing and payouts. One Next.js codebase serves web, iOS and Android through Capacitor.

- Stripe Connect payments: payout onboarding, seat and usage-metered billing, per-invoice checkout links with fee splits, and idempotent webhook reconciliation.
- Two-way Google and Apple calendar sync with encrypted OAuth tokens and a retryable job queue; realtime crew chat per event on Supabase Realtime.

Next.js, React, Fastify, tRPC, Prisma, Supabase, Stripe Connect, Capacitor

Conek, for A-SAFE Digital asafe.com | case study **Software Engineer I**

Quote-to-cash platform for A-SAFE, integrated with their CRM, drawing and routing tools.

- Maintained and scaled event-driven microservices on NATS JetStream with type-safe contracts and clean architecture layering; inter-service latency under 100ms.
- Wrote 500+ unit and integration tests holding 95% coverage.

TypeScript, Node.js, Fastify, NATS JetStream, Next.js, PostgreSQL, AWS

Sportefy sportefypakistan.com | App Store | Play Store | case study **Trainee Software Engineer**

Sports booking and matchmaking app live in Pakistan: a Flutter app on the App Store and Play Store, a Next.js admin panel and a NestJS API.

- Flutter app with BLoC and clean architecture: QR check-in, wallet checkout, realtime match chat, an offline request queue with replay and a server-driven update gate.
- NestJS API and Next.js admin dashboard on a versioned, additive-only REST contract with OpenAPI drift checks in CI; no release broke older installs.
- Payments through Paymob behind a pluggable gateway adapter, with a wallet and credits ledger, promo codes, and role-based access from one shared permission matrix.

Flutter, NestJS, Next.js, Prisma, PostgreSQL, Supabase, Paymob

PROJECTS

RiskSense, Loan-Default MLOps Pipeline github **Final-Year Project with Meezan Bank Limited**

ZenML training pipeline with MLflow experiment tracking that predicts loan default on 39,503 distributor-financing records: derived default labels and repayment features, KNN imputation for missing KIBOR tenures, class weighting.

Python, ZenML, MLflow, scikit-learn, XGBoost, LightGBM, CatBoost

Automated Crypto-Derivatives Trading System

Event-driven TypeScript monorepo: strategy processes publish signals over Redis Streams to one execution service, the only component that holds exchange keys and places orders; every trading limit sits in one pure risk function.

TypeScript, Node.js, Fastify, tRPC, Next.js, PostgreSQL, Redis Streams

Voice AI Patient-Intake Agent github

Phone agent that registers new patients over a U.S. number in English or Spanish: a Retell LLM agent with tool-calling functions that saves to Postgres only after the caller confirms a read-back.

TypeScript, Retell AI, OpenAI, Fastify, Zod, PostgreSQL, Next.js

Faculty Reviews, Course Review Platform github **University team project, 2024**

Course and instructor reviews for IBA students: a Flutter app, a React web app and a Node.js, Express and MongoDB API. Wrote the API, the Flutter app and most of the web app in a three-person team.

Flutter, React, Node.js, Express, MongoDB

EDUCATION

Institute of Business Administration (IBA), Karachi **Graduated 2025**
BS Computer Science. National Talent Hunt Program Scholar (top 33 of 7,290 candidates).

ACHIEVEMENTS

- **Teaching Assistant, Mobile Application Development (IBA):** mentored 100+ students; class assessment averages rose 30%.
- **Ignite Hackathon:** 1st place in a 3-hour challenge to think up and build something unique.
- **Trading Bazaar:** 1st place for algorithmic trading strategies.
- **Web3 Security Research:** 4 audit submissions in review on Cantina and Immunefi (Symbiotic, Term Finance).
- **Autonomous Maze-Solving Robot:** 3rd of 45 teams, built in embedded C and again in pure discrete logic.